COURSE
OVERVIEW

MECHANICAL LIFTING

The Mechanical Lifting course is designed to provide learners with the knowledge and practical awareness required to safely perform lifting activities using basic mechanical lifting equipment.

The course focuses on planning lifting tasks, assessing loads, selecting suitable lifting aids, and applying controlled lifting techniques. Emphasis is placed on hazard identification, load stability, centre of gravity, communication during lifting operations, and the use of exclusion zones to reduce the risk of injury, equipment damage, and uncontrolled load movement.

BY COMPLETING THIS COURSE, LEARNERS WILL BE ABLE TO:

- Identify hazards associated with mechanical lifting activities
- Assess loads and plan lifting tasks safely
- Select and use basic mechanical lifting equipment correctly
- Apply safe lifting techniques and correct posture
- Maintain exclusion zones during lifting operations
- Communicate effectively to ensure lifting activities are carried out safely and in a controlled manner

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GRAVITY VALUES

RELATIONSHIPS

Creating and maintaining meaningful relationships with all stakeholders of Gravity.

INTEGRITY

Conducting day-to-day business with integrity at all times.

SOLUTIONS **DRIVEN**

Being solution driven.

QUALITY & **EXCELLENCE**

Striving for excellence and constant improvement.

WELCOME, LEARNER!

Working at height is inherently risky — and this includes training exercises. The best way to stay safe is by learning and practising the correct techniques. This manual will help you:

- Understand the basics of fall arrest and basic rescue
- Learn how to identify and manage height-related hazards
- Build the skills needed to work safely and confidently

Working at height means doing any job where you could fall from one level to another. This includes tasks done:

- Above the ground or floor
- Near open edges
- Near fragile surfaces
- Next to holes or gaps in the ground or floor
- Close to excavations

Note: Work at height does not include slipping or tripping hazards on the same level.

HOW TO USE THIS MANUAL?

This guide supports the Fall Arrest and Basic Rescue Course and is meant to be used with the help of a trained instructor.

At Gravity Training, all facilitators are highly trained to deliver top-quality learning experiences. The content has been designed to be clear, easy to remember, and practical for real worksites.

SITE ESTABLISHMENT

Before any work begins, the site must be safe.

Use this simple checklist

01

Follow the Rules:

Make sure you follow all client rules.

Example: If night work is not allowed, do not work at night.

02

Rescue Plan

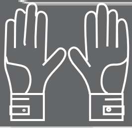
Have a rescue plan ready. A rescue kit and first aid kit must be on site.

03

Work in Pairs

There must be at least two workers on site.

If something goes wrong, one can help or call for help.



04

Check Equipment

All tools and gear must be safe and working before use.

05

Safe Site

Check the building or structure.

Make sure it is strong and safe to work on.

06

Manage Traffic

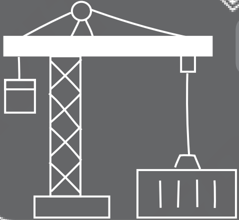
If there are cars or trucks nearby, there must be a traffic safety plan.

07

Do a Risk Assessment

Check what dangers there are.

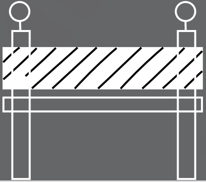
All workers must understand the risks before starting work.



08

Use the Right Safety Gear

Make sure the fall protection fits the job and the site.



EXCLUSION ZONES **& BARRICADES**

Exclusion Zone: An area where no one is allowed to walk or work.

Example: A 2-metre space around a hole in the roof.

Barricade: A physical barrier to block people from danger.

Example: A fence or tape to stop people from falling.

SITE SAFETY

Be healthy and ready before starting work.

Only do jobs you are trained and allowed to do.

Never work under the influence of alcohol or drugs.

Always wear your safety gear (PPE, harness if working at height).

Use harnesses correctly and stay connected at heights.

Never work alone and always follow safety steps.

GRAVITY

SITE SAFETY

- **Keep people away** from areas below when working above.
- **Don’t walk under lifted loads** or stand in dangerous areas.
- **Only trained people** can do electrical, underground, or confined space work.
- **Follow road rules.** Wear helmets. No passengers on open trucks..
- **Don’t drive tired** or distracted.
- **Stop unsafe work!** Everyone has the right to speak up.

OCCUPATIONAL HEALTH AND SAFETY ACT, 1993
DRIVEN MACHINERY REGULATIONS, 2015

The Minister of Labour has, under section 43 of the Occupational Health and Safety Act, 1993 (Act No. 85 of 1993), after consultation with the Advisory Council for Occupational Health and Safety, made the regulations in the Schedule.

Definitions

1. In these Regulations, "the Act" means the Occupational Health and Safety Act, 1993 (Act No. 85 of 1993), and any word or expression to which a meaning has been assigned in the Act shall have the meaning so assigned, and, unless the context otherwise indicates -

"**Block and tackle**" means a lifting device consisting of one or more pulley blocks reeved with fibre ropes, used solely for the raising and lowering of a load or for moving it horizontally, but does not include chain blocks, lever hoists or steel- wire
rope pullers;

"**Capstan -type hoist**" means a rotating machine used to control or to apply force to move or raise loads by traction on a rope or cable; Wich can be remotly operated and has A intergrated bracking system.

"**Competent person**" means a person who has the knowledge, training, experience and qualifications specific to the work performed: provided that where appropriate qualifications and training are registered in terms of the provisions of the South African Qualifications Authority Act, 1995, those qualifications and that training shall be deemed to be the required qualifications and training;

"**Hand- powered lifting device**" means a lifting device consisting of one or more sheave components reeved with chains, steel rope or fibre ropes, used solely for the raising and lowering of a load or for moving it horizontally and includes chain blocks, lever hoists, hand chain hoists, steel -wire rope pullers and hoists, but does not include hand -powered hydraulic lifting devices;

"**Lifting machine**" means a power- driven machine that is designed and constructed for the purpose of raising or lowering a load or moving it in suspension, but does not include an elevator, escalator or hand- powered lifting device;

"**Lift truck**" means a mobile lifting machine, but does not include -
(a) a vehicle designed solely for the purpose of lifting or towing another vehicle;
(b) a mobile earth -moving machine; or
(c) a vehicle designed solely for the removal of a waste bin;

"**Lifting machinery entity**" means a legal entity approved and registered by the chief inspector in terms of regulation 19;

"**Lifting machinery inspector**" means a person who is employed by a Lifting Machinery Entity and who is registered by the Engineering Council of South Africa in terms of the Engineering Profession Act, 2000 (Act No. 46 of 2000);

"**Lifting tackle**" means chain slings, wire rope slings, woven webbing slings, master links, hooks, shackles and swivels, eye bolts, lifting or spreader beams, tongs, ladles, coil lifters, plate lifting clamps and drum lifting clamps used to attach a load to a lifting machine;

"**load path**" means all the parts of the lifting machine under stress during the lifting operation;

"**man- cage**" means a platform enclosed on all sides, whether closed or open at the top, designed for the purpose of raising and lowering persons by means of a lifting machine, but does not include mobile elevated work platforms and suspended access platforms;

"**point of operation**" means that place in a machine where material is positioned and where the actual work is performed;

"**safe working load**" means the mass load applicable to a piece of equipment or system as determined by a competent person taking into account the environment and operating conditions;

"**thorough examination**" means examination or inspection to determine whether the equipment is safe to use;

"**training provider**" means a training provider for lifting machinery operators approved and registered by the chief inspector in terms of regulation 20;

Scope of application
2. These Regulations shall apply to the design, manufacture, operation, repair, modification, maintenance, inspection, testing and commissioning of driven machinery, lifting machines, hand -powered lifting devices and lifting tackle

18.(1) No user may use or permit the use of a lifting machine or hand- powered lifting device unless -
(a) it has been designed and constructed in accordance with a generally accepted technical standard;
(b) it is conspicuously and clearly marked with the safe working load: provided that when such safe working load varies with the conditions of use of the manufacturer, a table showing the safe working load with regard to every variable condition shall be posted by the user in a conspicuous place easily visible to the operator;
(c) the manufacturer's identification plate displaying the name of the manufacturer, the design standard, the serial or reference number and the country of origin is affixed to such machine; and

(d) it has at all times at least three full turns of rope on the drum of each winch that forms part of such a machine when such winch has been run to its lowest limit, and that is controlled by an automatic cut -out device: provided that paragraphs (b) and (d) above shall not apply to capstan -type hoists.

(2) The user shall ensure that every power- driven lifting machine is fitted with a brake or other device capable of holding the safe working load should -
(a) the power supply or lifting effort fail;
(b) the load attachment point of the power- driven lifting machine reach its highest and lowest safe position; or
(c) the load condition be greater than the rated load condition of that machine.

(3) The user shall cause every chain or rope that forms part of the load path of a lifting machine or hand- powered lifting device to have the factor of safety prescribed by the standard to which that machine was manufactured: provided that in the absence of such prescribed factor of safety, chains, steel -wire ropes and fibre ropes shall have a factor of safety of at least four, five and 10, respectively, with regard to the safe working load of that machine.

(4) The user shall cause every hook or any other load -attaching device that forms part of the load path of a lifting machine or hand -powered lifting device to be so designed or proportioned that accidental disconnection of the load under working conditions cannot take place.

(5)(a) The user shall cause the entire installation and all working parts of every lifting machine or hand -powered lifting device, as well as ancillary lifting equipment used with the machine or device, excluding lifting tackle, to be subjected to a thorough examination and a performance test, as prescribed by the standard to which the lifting machine was manufactured, by a lifting machinery inspector of a lifting machinery entity, which shall determine the serviceability of the structures, ropes, machinery and safety devices before they are put into use and every time they are dismantled and re- erected, and thereafter at intervals not exceeding 12 months: provided that, in the absence of a manufacturing standard or a standard incorporated under section 44(1) of the Act, the whole installation of the lifting machine shall be tested with 110% of the safe working load applied over the complete lifting range of such machine and in such a manner that every part of the installation is stressed accordingly.

(b) The lifting machinery inspector of the lifting machinery entity referred to in paragraph (a) must have knowledge of the erection, load- testing and maintenance of the type of lifting machine or similar machinery involved.

(c) Notwithstanding paragraph (a), mobile cranes, self -erecting cranes and mobile elevated work platforms shall be excluded from the performance test after each re- deployment within the 12 -month period referred to in that paragraph.

(6) Notwithstanding sub-regulation (5), the user shall cause all ropes, chains, hooks or other attaching devices, sheaves, brakes and safety devices forming an integral part of a lifting machine or hand -powered lifting device to be subjected to a thorough examination by a competent person at intervals not exceeding six months.

(7) (a) Every user of a lifting machine or hand -powered lifting device shall at all times keep on their premises a register in which the user shall record or cause to be recorded full particulars of any performance test and examination referred to in sub-regulations (5) and (6) and any modification or repair to such lifting machine or hand- powered lifting device, and shall ensure that the register is available on request for inspection by an inspector.

(b) Every user of a leased lifting machine or hand -powered lifting device shall at all times keep on their premises a register in which the user shall have the latest applicable performance test and service records not older than 12 months.

(c) The owner and the lessor of leased equipment shall keep and maintain full service history records on their premises for at least 10 years.

(8) No user shall require or permit any person to be moved or supported by means of a lifting machine unless that machine is fitted with a man -cage designed and manufactured according to an approved SANS standard approved for that purpose by an inspector and after a risk assessment has been done.

(9) No user shall use or permit any person to use any power- driven lifting machine unless it is provided with -
(a) in the case of a power -driven lifting machine with a lifting capacity of greater than 5 000 kg, a load indicator capable of indicating to the operator of the machine the mass of the load being lifted: provided that such device shall not require manual adjustment, from the application of the load to the power- driven lifting machine until the release of that load, using any motion or combination of motions permitted by the crane manufacturer to ensure safe lifting; and /or
(b) a load -limiting device that will automatically arrest the driving effort whenever the load being lifted is greater than the safe working load of the power -driven lifting machine at that particular radius, using any motion or combination of motions permitted by the crane manufacturer to ensure safe lifting: provided that such device shall not arrest the driving effort when the power- driven lifting machine is being operated into a safer position: provided that power- driven lifting machines manufactured or refurbished prior to the commencement of these Regulations shall be deemed to comply with these Regulations.

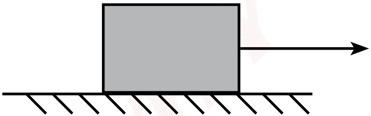
(10) No user may use or allow the use of any lifting tackle unless -
(a) every item of lifting tackle is well constructed of sound material, is strong enough, is free from defects and is constructed in accordance with a generally accepted technical standard;
(b) every lifting assembly consisting of different items of lifting tackle is conspicuously and clearly marked with traceable identification particulars and the safe working load that it is designed to lift with safety;
(c) the ropes, chains or woven webbing have a factor of safety with respect to the safe working load they are designed to lift; the safety factor being
(í) 10 for natural -fibre ropes;
(ii) seven for man -made fibre ropes or woven webbing;
(iii) six for steel -wire ropes, except for double -part spliced endless sling legs and double -part endless grommet sling legs made from steel -wire rope, in which case the factor of safety shall be at least eight;
(iv) five for steel chains; and

(v) four for high- tensile or alloy steel chains: provided that when the load is equally shared by two or more ropes or chains the factor of safety may be calculated in accordance with the sum of the breaking strengths taking into consideration the angle of loading;
(d) all lifting tackle is inspected and discarded if such items show any sign of damage, defect, wear or distortion that would make them unsafe for use, as per manufacturer's specification; and
(e) such lifting tackle is examined at intervals not exceeding three months by a competent person, appointed by the user in writing for this purpose, who shall record and sign results of such examination.

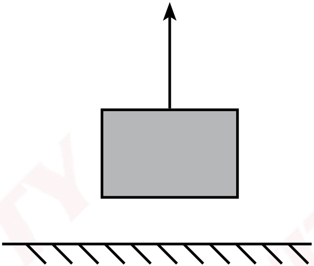
(11) **The user shall ensure that every lifting machine is operated by an operator specifically trained for that particular type of lifting machine:** provided that in the case of a lifting machine listed in the National Code of Practice for Training Providers of Lifting Machine Operators, the user shall not require or permit any person to operate such a lifting machine unless the operator is in possession of a certificate of training, issued by a training provider accredited by the Transport Seta approved for the purpose by the chief inspector.

A. Lifting vs Pulling

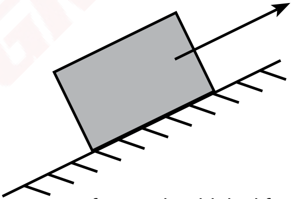
Consider all these for lifting plan.



Pulling = should the pulling method fail, the load remains stationary.



Lifting = should the lifting method fail, the load will fall.



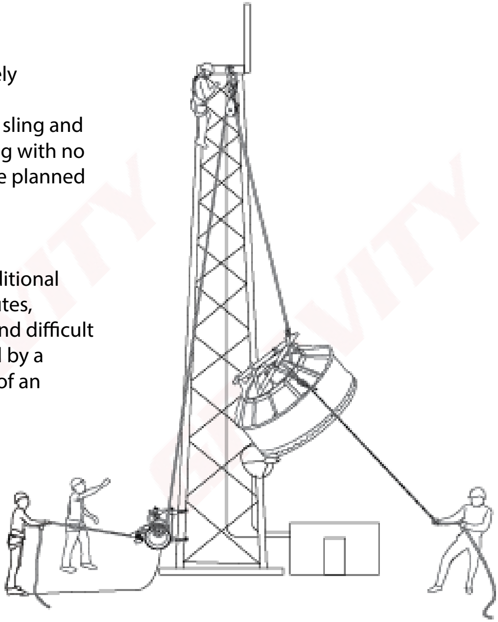
Lifting = should the lifting method fail, the load will slide downwards.

B. Routine v Non-routine lift

Lifts are typically divided into two categories, namely

1. Routine:
   A simple lift done frequently involving simple sling and anchoring as well as loads that are easy to sling with no horizontal components. Lifts like these may be planned by a person competent in lifting.

2. Non-routine:
   A lift that is not done frequently involving additional hazards such as difficult to navigate lifting routes, complex anchoring, multiple lifting systems and difficult to sling loads. Lifts like these must be planned by a person competent in lifting with the support of an engineer



C. Safety factors

DIFFERENCES BETWEEN LIFTING EQUIPMENT, AND ROPE ACCESS & FALL ARREST EQUIPMENT:

Lifting equipment & safety factors:

- Testing certificates issued annually by an LMI.
- Lifting equipment will always have a W.L.L. (Working Load Limit) that indicates the maximum load that can be lifted with the equipment.
- The rating will be in TON or KG.
- Lifting equipment also has a safety rating, for example 7:1, which implies that a 500 kg WLL rope will break at 3.5 ton. This will be indicated on the equipment.
- A lifting machine has a safety factor of 5:1 unlike a pulling machine which has a 3:1 safety factor.
- Ensure that the correct equipment is used to ensure that a lift is as secure as possible during a lift.

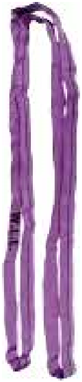
Rope access & fall arrest equipment, and safety factors:

- Equipment used primarily for the support and protection of workers at height, but may also be used for Rope Rigging
- Equipment inspections done at least every 6 months by a competent equipment controller.
- Rope Access and Fall Arrest equipment will have a S.W.L (Safe Working Load) on the equipment that will indicate the amount of weight (in KN; 1 KN = 100 kg) that can safely be applied onto the equipment.
- However, the M.B.S. (Minimum Breaking Strain) of equipment indicates the amount of weight the equipment can hold before it will break.
- In Rope Access and Fall arrest equipment the relation between the S.W.L. and the M.B.S. will be the safety rating (not indicated on the equipment) and will always be 10:1.

D. Rigging and lifting equipment

1. Webbing lifting slings (EN 1492-1):

- Webbing Lifting Slings are coded per lifting rating: e.g. green is for 2 ton WLL etc.
- Webbing slings have a safety rating of 6:1 (unless otherwise stated by the manufacturer) which means that a 1 T sling will break at 6 T.
- Colour coded as per EN 1492-1.
- Lifting tackle must be inspected every 3 months by A competent person rever to DMR section 10(e)
- User must conduct pre-use inspections as per software requirements.

	Colour coded according to DIN-EN 1492-1	Working Load Limits with 1 webbing sling						
		Straight Lift	Choked Lift	Basket 0 Degrees	Basket 0-7 Degrees	Basket 7-45 Degrees	Basket 45-60 Degrees	Basket 45-60 Degrees
								
		1.0	0.8	2.0	1.4	1.0	0.7	0.5
	WLL 1T	1000	800	2000	1400	1000	700	500
	WLL 2T	2000	1600	4000	2800	2000	1400	1000
	WLL 3T	3000	2400	6000	4200	3000	2100	1500
	WLL 4T	4000	3200	8000	5600	4000	2800	2000
	WLL 5T	5000	4000	10000	7000	5000	3500	2500
	WLL 6T	6000	4800	12000	8400	6000	4200	3000
	WLL 8T	8000	6400	16000	11200	8000	5600	4000
	WLL 10T	10000	8000	20000	14000	10000	7000	5000
	WLL 12T	12000	9600	24000	16800	12000	8400	6000

Note: When changing the configuration of the slings the WLL will either increase or decrease the strength of the sling.

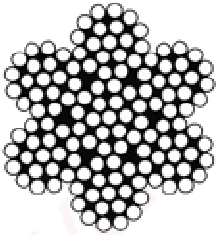
2. Wire rope slings (SANS 7531):

- Wire rope slings are used where sharp edges may damage webbing or rope slings.
- Wire rope slings are very strong and can range from a breaking strength from 45kN to 4000kN
- Both ends must be clipped with a connector.
- Lifting tackle must be inspected every 3 months by A competent person rever to DMR section 10(e).
- User must conduct pre-use inspections as per hardware requirements.

3. Steel wire rope (ISO 17893):

- Wire ropes are made up off groups of twisted wire strands (cable) with a rope diameter from 10mm to 32mm, the safety factor for wire slings are 6:1. That means that a 1T wire sling will break at 6T.
- Steel wire and steel wire slings are made up of wires that are wound together around a King Wire to form a Strand. Strands are then wound together around a core to form a wire rope.
- Each wire rope must have a certificate of conformity.
- Lifting tackle must be inspected every 3 months by A competent person rever to DMR section 10(e).

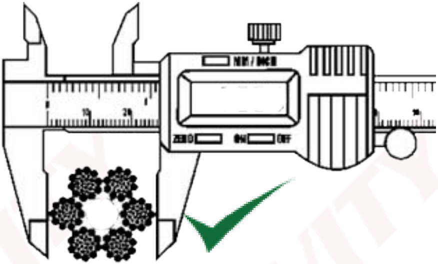
Fibre Core

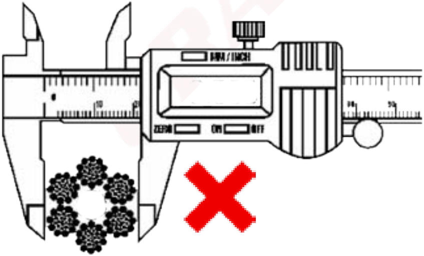


Wire Core

How to measure steel wire rope

The correct way to measure the diameter of steel wire rope is to place the calliper over each pair of opposite strands. Three readings will be taken on a six strand rope as per example.





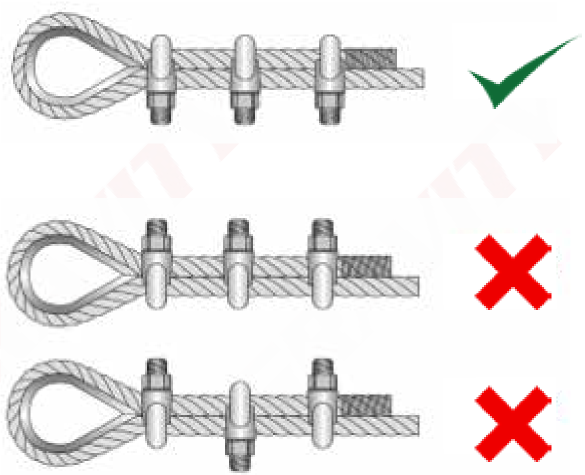
Clamping a wire rope

Always ensure that the amount of rope turned back from the thimble or loop is enough to accommodate all of the required clamps.

Ensure that the clamps or saddles are orientated in such a way that the saddle is not on the dead (turned back) rope.

NEVER SADDLE A DEAD HORSE

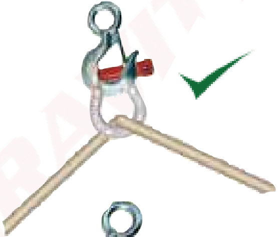
Although there are other end terminations of wire ropes, no end termination may decrease a wire rope by more than 20%.



4. Shackles (EN 13889:2003):

- Bow shackles are available from 0.33 Ton up to 120 Ton.
- The most common shackle is the Screw Pin Bow Shackle.
- Must be inspected annually by an lifting machinery inspector (LMI).
- User must conduct pre-use inspections as per hardware requirements.

Advantages of bow shackles:



1. The advantage of a bow shackle above the D-shackle is the fact that due to its shape it can accommodate more than 1 attachment.
2. In environments where vibration is a problem or more permanent attachment is needed, the Bow Shackle is a better choice than a Karabiner or Maillon.

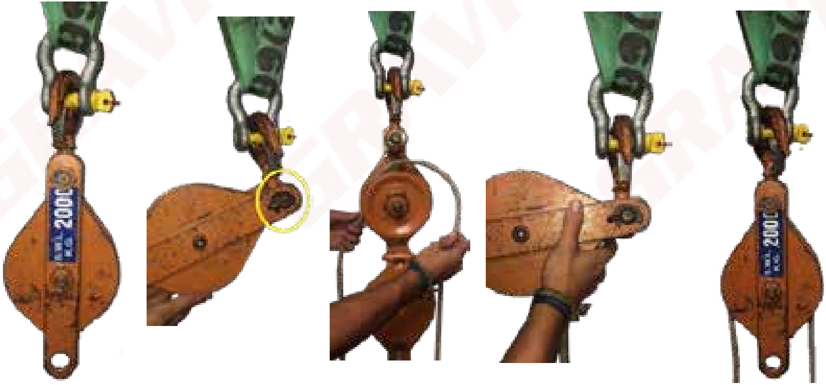
Safe practice (uses) of bow shackles:

1. Ensure that ONLY rated shackles are used and not shackles that are available from retail shops and that are used for decoration etc.
2. Ensure that with Bow Shackles with safety pins, the split pin is inserted and properly split and bent open to ensure no accidental undoing of the nut is possible.

5. Pulleys and snatch blocks:

- Pulleys and snatch blocks are used to reduce friction in a system as well as changing the direction of pull in the system
- Pulleys to be used must be certified with a working load limit
- Snatch blocks have bigger sheaves and higher working load limits

Installing a rope in a snatch block



6. Chain slings:

- Chain links are available from 1 Ton up to 100 Ton.
- Chain links also vary in number of legs which affects the W.L.L.
- Must be inspected annually by a lifting machinery inspector (LMI).
- User must conduct pre-use inspections as per hardware requirements.

Use of chain slings:

1. Never make knots in the chain.
2. Never exceed 120° between the chain legs.
3. Make sure that the load is evenly distributed between all legs.
4. All legs not in use should be hooked back on the oval to avoid swinging or snagging.
5. Always refer to manufacturer's recommendations for safe use instructions

7. Eyebolts

- Eyebolts are available from 0.7 Ton up to 8.6 Ton.
- Eyebolts are only to be connected to points that are specified by the manufacturer.
- Must be inspected annually by a lifting machinery inspector (LMI).
- User must conduct pre-use inspections as per hardware requirements.

Use of eyebolts:

1. Store and handle eye bolts correctly.
2. Inspect eye bolts before and after use.
3. Ensure the eyebolt is suitable for the task at hand.
4. Ensure that the eye bolt and connection points are compatible and strong enough for the imposed load.
5. Ensure that the collar is fully seated when hand tight.
6. Ensure that load direction of bolts are correct and accounted for in the lifting plan.
7. Always refer to manufacturer's recommendations for safe use instructions

8. Gravity 500 Capstan hoist

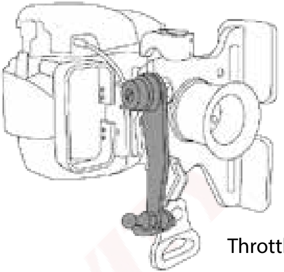
Suitable for many applications, capstan hoists are either electrically or gasoline powered. Always ensure that the Hoist has a sufficient WLL for the intended load.

Capstan Hoist used for lifting need to have a rope grab device on the lazy end of the drum to prevent a load from falling should the user let go of the rope.

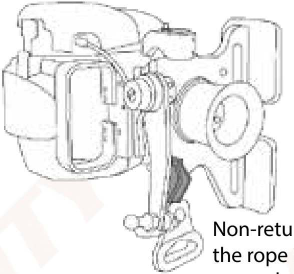
All hoists must be inspected by an LMI at least once a year.

The hoist should receive a minor service after every 2 site builds (+/-50 hours) and should receive a major service after 4 site builds (+/-100 hours).

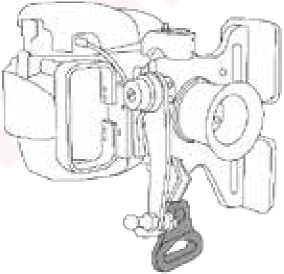
Pre-use checks:
Ensure all nuts and bolts are present and tight.



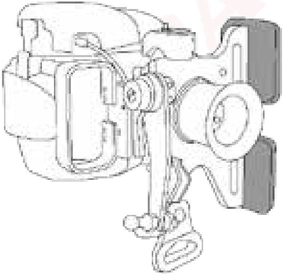
Throttle arm.



Non-return catch - ensure the rope is installed correctly.



Anchor point - ensure that the winch is anchored securely and that all the force passes through the bottom anchor. Ensure the anchor point is NOT connected with a metal connector.



Non-skid pads - ensure that the winch is orientated correctly for the intended lift.

Gravity 1200 Capstan hoist

This hoist has the same functions as the Eder 500, however this hoist is rated for slightly heavier loads and has a 2 stroke motor.

This hoist should be operated in a horizontal orientation and not in a vertical orientation like the Eder 500 due to anchoring difficulties.

2 Stroke pre mix chart

	Low quality OIL	Recomended quality OIL
	33:1	50:1
Fuel (L)	Oil (ML)	
5	152	100
7.5	227	150
10	303	200

When anchoring this hoist your are not allowed to anchor it on the frame, It has a section specifically for anchoring.

Pre-use checks:
All the same inspection criteria must be followed for this hoist as for the Eder 500.

9. Hoist compatible rope

Gravity Gear semi-static ropes (EN 1891)

Used throughout work at height systems and is a crucial part of working at height safety.

- the rope stretches between 5% and 10%.
- Is of kermantle construction and consists of two parts:
  Kern (Core) = 80% of the strength.
  Mantle (sheath) = 20% of the strength.

- S.W.L. = 300 kg
- M.B.S. = 3000 kg

The S.W.L will change depending on the knot configuration on the rope.

PLEASE NOTE THAT ONLY **GRAVITY GEAR** 11mm ROPES MAY BE USED WITH BOTH THE EDER 500 AND EDER 1200 HOIST**.**

10. Tirfor (Wire rope hoist)

Suitable for many applications, tirfors are lever operated hoists using a separate wire rope. They are one-man operated, using a telescopic operating handle, and they can work in any position and over any height of lift. They can replace conventional winches and other hoists for many applications.

Never operate the lifting and lowering levers simultaneously as this might result in releasing the load.
Do not use an extension lever for extra advantage as this might overload the device.

Avoid lifting less than 10% of the device's capacity and never lift less than 5% of the device's capacity as this might prevent the device's friction brake from activating.

Pre-use checks:
Ensure the shear pin is present and in good condition.
Ensure the clutch engages properly.

11. Lever hoist

Lever hoists are versatile manually operated hoists used for a variety of lifting, pulling, tensioning and material handling type applications.

Lever hoists operate by levering the handle to and fro to lift or lower the load. Lever hoists can be used at any angle and one of the main differences between a chain block and a lever hoist is that lever hoists have a neutral mode which enables the chain to free wheel through the hoist for the purpose of taking up slack chain quickly.

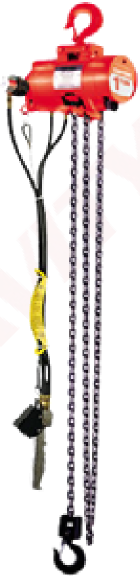
The length of the lift is limited to the length of chain.

Avoid lifting less than 10% of the device's capacity and never lift less than 5% of the device's capacity as this might prevent the device's friction brake from activating.

Pre-use checks:
Ensure chain moves smoothly in both directions.
Ensure the clutch engages properly.

12. Chain hoist/Air hoist

Manual chain hoists are commonly known as chain blocks and are designed for heavy duty lifting and material handling operations. Chain blocks raise and lower loads by pulling on the hand chain. Chain blocks can be of single or double fall configurations depending on the capacity of the block.



Air hoists are similar to chain hoists but are pneumatically powered. This provides an advantage over chain hoists as this allows them to be operated remotely.

The length of the lift is limited to the length of chain.

Avoid lifting less than 10% of the device’s capacity and never lift less than 5% of the device’s capacity as this might prevent the device’s friction brake from activating.

Pre-use checks:
Ensure chain moves smoothly in both directions.
Ensure the clutch engages properly.

13. Gin poles/Derricks

Gin poles are used to establish anchor points away from the main structure to help prevent possible damage to the structure or the load as well as reduces the need for taglines. Gin poles also aids in spreading the load over a larger surface on the tower.

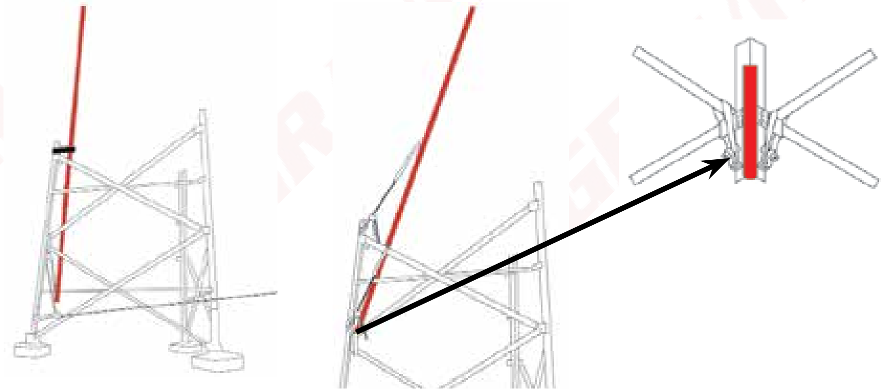
However, it is of utmost importance that the anchor points to which the gin pole are connected and the gin pole itself are strong enough for the imposed loads and that the installation procedure as given by the manufacturer is followed. This information may be obtained from the owner of the structure and/or the manufacturer of the gin pole, these requirements must be adhered to at all times!

Derricks may also be used for similar application, However the Derrick is a slightly more advanced system that allows one to move a suspended load in horizontal direction, side to side .

Installing the Gin pole

When the first legs are erected and the first level of bracing has been installed, the gin pole needs to be installed in order to facilitate the lifting and installation of the subsequent sections.

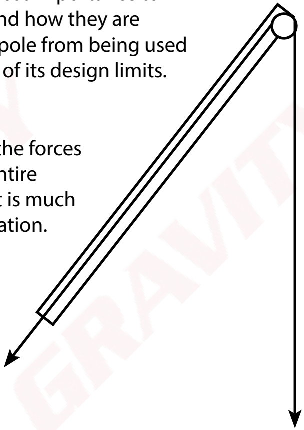
- Two tag lines are to be connected to the gin pole in order to guide the pole into place.
- The base of the pole is placed in the corner of the tower. Which will serve as a pivot point for the gin pole.
- When the gin pole is upright, a worker is to connect a chain hoist (lever hoist) to a anchor point on the pole in order to stabilise and secure the top of the pole to the structure.



Construction forces

When using a gin pole it is of utmost importance to be aware of the forces involved and how they are applied. This will prevent the gin pole from being used incorrectly or being used outside of its design limits.

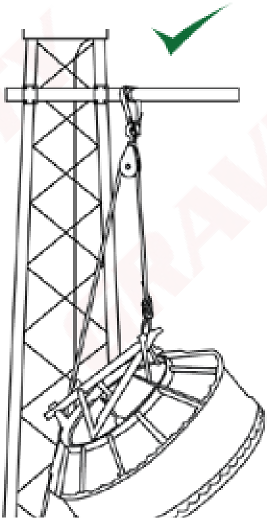
When the gin pole is vertical the forces are distributed through the entire length of the pole, meaning it is much stronger in the vertical orientation.



When the gin pole is tilted or angled away from the vertical orientation, the force applied to the gin pole and the forces experienced at the base increase dramatically as well as increasing the bending forces within the pole which can cause the gin pole to deform or even cause it to fail.

Typically the certified gin pole may not be angled more than 45° and may not be loaded with more than 220kg, however the manufacturer will need to supply you with the angles and Working load limits.

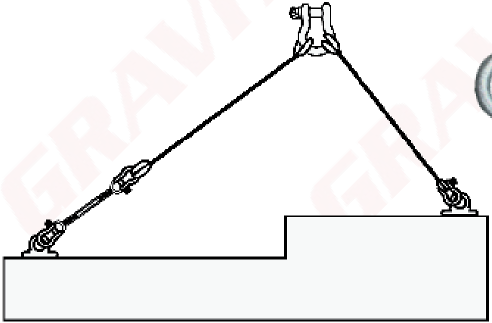
When using gin poles in a horizontal orientation, be aware that the further out you rig on the pole, the more force will be put on the tower, ensure that the gin pole and tower are strong enough for the imposed loads and that the gin pole is anchored in such a way that it will not bend, twist or crack. Always ensure that the gin pole is signed off and is used within its limits as determined by the manufacturer.

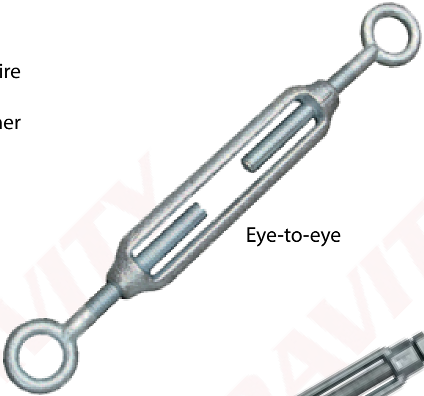


14. Turnbuckles

Turnbuckles are used to either tension a steel wire rope or to balance a lift when using wire slings

- All turnbuckles must have closed ends, either eye-to-eye or jaw-to-jaw.





Eye-to-eye

Jaw-to-jaw

15. Beam clamps

A beam clamp is a useful maintenance tool that enables a simple fast, portable and secure means of attaching a lifting point to a beam.

- Do not use beam clamps on girder widths that are not within the clamps stipulated grip range.
- Ensure that the clamp is securely fitted to the beam and that the centre line of the lifting point is aligned with the centre web of the beam.
- If two clamps are to be used in a dual lift situation a spreader bar will be required. The load must be lifted slowly and care must be taken to ensure that a full load in excess of the WLL of a single clamp is not put on any one clamp. A careful, equal lift should ensure the load is shared equally between both clamps.
- If a clamp is to be used to suspend a pulley, the additional loading caused by the downward pull must be considered when selecting the WLL.
- Select clamps which are compatible with the dimensions of the beam and which are in excess of the WLL of the beam.

16. Spreader/Equalizing beams

Spreader or equalizing beams are used to distribute the loads effectively over two anchor points whilst keeping the direction of pull on the load anchor points at 90°. This evenly distributes the weight of the load across the two slings, which then connect to a crane, hoist, or other lifting machine. Two lugs on the bottom (one at each end) connect to a sling or hook which are then connected to the load.

Remember that spreader beams require more freeboard in order to accommodate the overhead slings.

17. Jacks

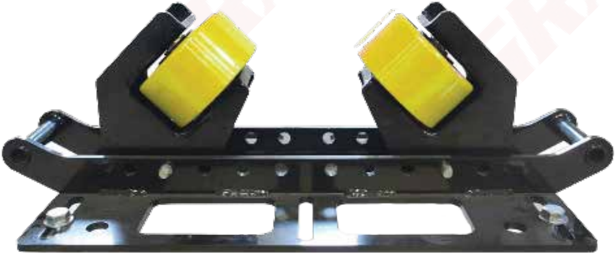
Hydraulic jacks may be used to lift heavy loads a small distance of the ground to allow the load to be slung appropriately without causing damage to the slinging equipment, similar to lifting blocks during rope rigging operations.

- Ensure that the jack is stable on the ground and cannot tip over once weight is applied to it.
- Ensure that the load is balanced so that it does not fall off the jack once lifted.

18. Sliding shoes and rollers

In order to manipulate loads over short distances on ground level, either sliding shoes or rollers may be used depending on the size and weight of the load.

- Sliding shoes are more appropriate for heavier larger loads.
- Rollers are more appropriate for long round loads.
- Hydraulic jacks may be used to lift loads onto rollers or sliding shoes.

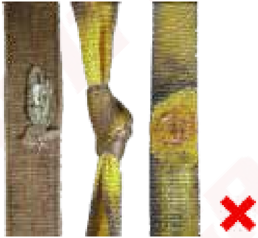


E. Equipment management
1. Pre-use inspections

Users must inspect all the equipment to be used before work starts. This must be done to ensure that all equipment is safe for use and fit for purpose in accordance with DMR 18

Equipment is divided into 2 categories namely hardware and software. Each of these have their own inspection criteria. Some pieces of equipment (e.g a harness) fall into both categories and must be inspected as such.

Software



Check for:

- Date of manufacture
- Condition of stitching
- Discolouration
- Cuts
- Burns
- Wear and tear
- Certification

Hardware

Check for:

- Rust
- Working action
- Movement
- Wear and tear
- Deformation
- Cracks

2. Lifting machinery inspector

The LMI must have the necessary knowledge on the user's legal responsibilities in terms of lifting tackle and hoists.

It is a legal requirement to inspect all lifting equipment at various mandatory intervals. A lifting machinery inspector shall be responsible for the inspection of lifting machinery **in periods not exceeding 12 months**. All lifting machinery is subjected to a visual inspection **every 6 months** by a competent person and lifting tackle shall be inspected by a trained and competent person **in periods not exceeding 3 months.** Furthermore, the user must record the results of all these inspections in a register, which must be kept for this purpose, and the register must be available for inspection, or auditing, at any time.

This applies to:

Chain slings, wire rope slings, webbing slings and components, such as shackles, hooks and eyebolts. Visual inspections of chain blocks and lever hoists, clamps and trolleys, electric and pneumatic hoists, lifting beams and wire rope winches

Unauthorized repairs or modifications of lifting equipment will invalidate the certification of the equipment

3. Housekeeping:

Whilst on site, it is important to ensure that the equipment being used is handled in such a way that they do not suffer any unnecessary damage or exposure to environmental conditions. When equipment is not in use , store them in their bags or bins. Ensure that equipment is moved out of the areas where workers move to prevent slips and trips or other accidents.

Failure to properly care for equipment can cause it to fail which can cause serious injury or death.

4. Storage:

- Gear must be stored in a cool, dry environment.
- Gear must be stored in allocated places according to work site procedures.
- Gear can safely be transported using PVC covered gear bags.
- Store in designated space or dedicated rope bag.

5. Lifespan:

Confirm the lifespan of equipment with the manufacturer's specifications. Lifetime is generally taken (unless otherwise specified by the manufacturer) from the date of production with the exception of metal components (connectors, braking devices, etc.) for which the lifetime is determined only by frequency of the use, wear and tear, and working action.

F. Anchor points

1. Anchor points for PPE

Anchor points for fall arrest and rope access are mainly divided into two categories during normal work at height activities namely:

- Self-identified anchor points, and
- Planned anchor points (EN 795:2012).

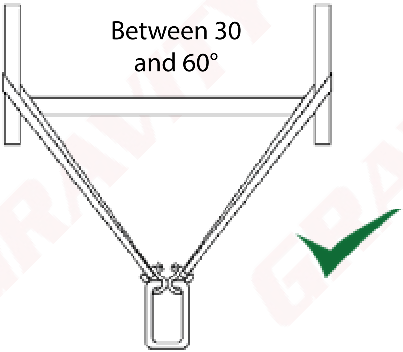
However, the loads during mechanical lifting may easily exceed these limits and as such it is required that the lifting supervisor obtain proof that the intended anchor points a suitable for the imposed loads.

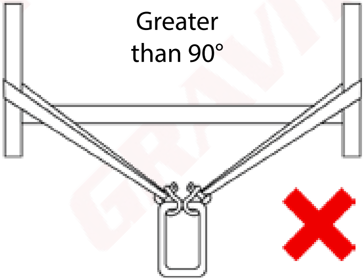
2. Anchor points for rigging

- Anchor points should be strong enough to withstand the forces generated by the rigging system.
- Use two separate anchor points where possible or when in doubt.
- Only use the vertical members of the tower or a gin pole attached to the vertical members to anchor a heavy lifting system to. Never anchor heavy systems to diagonal or horizontal members on a tower.
- All anchor points approved for intended loads by a competent person.

3. Load sharing anchors (LSA)

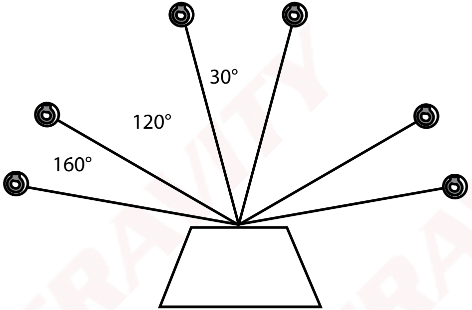
As far as possible, it is always a good idea to share the load between two SEPARATE anchor points. However, it is important to remember that angles play a big part when sharing loads, and angles should be kept to a MINIMUM.

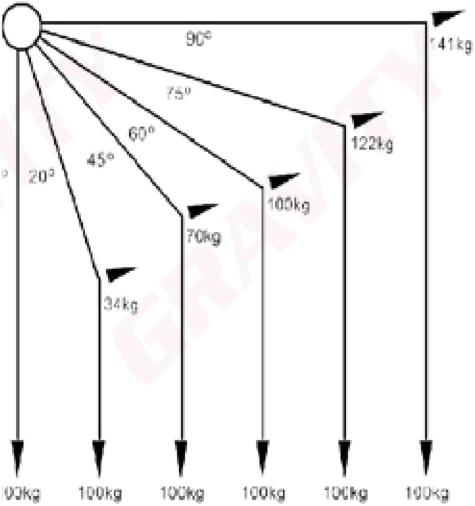




4. Angles and forces

When rigging, the angle between anchor points determines the force experienced by each anchor. Below is a breakdown of the angles and their corresponding forces.





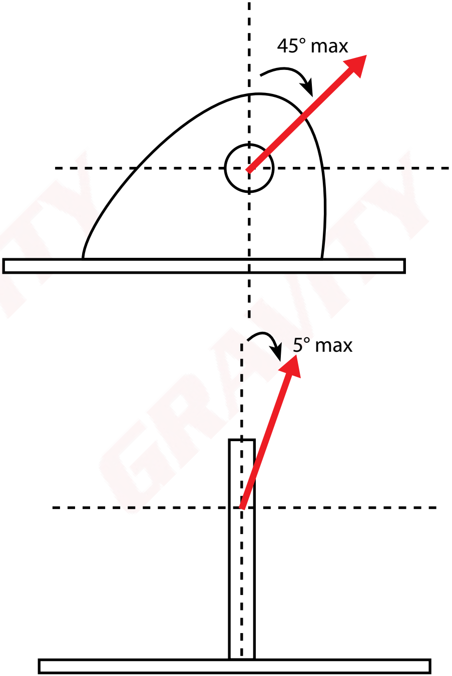
Angle	Force experienced by each anchor
0°	50% of load
30°	52% of load
120° (dangerous)	100% of load
160° (very dangerous)	300% of load

5. Lifting with pad eyes

Pad eyes or lifting lugs are of very specialized design. The design must be done by engineers and also be signed off by an engineer after installation.

- It is good practice to assign a lug number to each lug – for the future identification of the lug (quality control measure).
- The centre hole on a lifting lug should never be flame cut. The lugs might need to undergo NDT (Non Destructive Testing) the flame cut might leave flaws in the material.
- In most instances, it is also a good idea to examine the structure the lug is connected to – as there is no use having a strong lug attached to weak structure.

Pad eyes must be used with in a 5° angle from the spine of the pad and never more than 45.



G. Lifting

1. Team composition

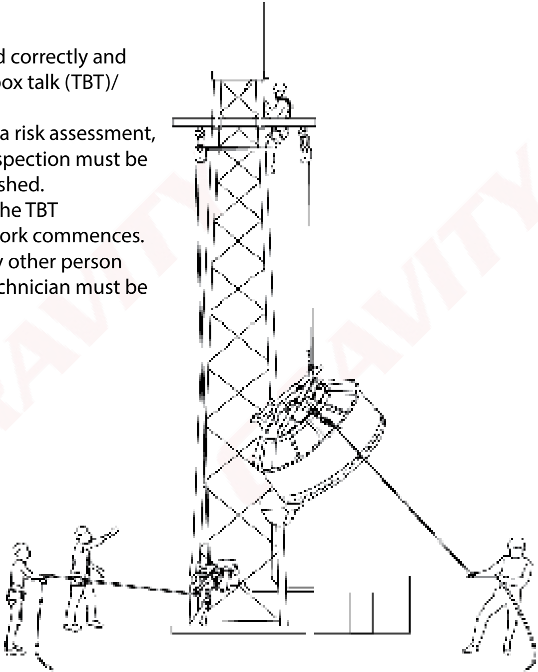
Before any mechanical lifting may commence on site, ensure that the following team members are on site with the correct documentation. Remember that any individual may refuse to work in any environment they deem unsafe or dangerous.

Team compositions based on each lifting scenario					
Team member	Scenario A 0 – 6kg	Scenario B 6 – 50kg	Scenario C 50 – 100kg	Scenario D 100 – 500kg	Scenario E >500kg
Rigging supervisor	✓	✓	✓	✓	✓
1st Basic rigger/fly line operator	✓	✓	✓		✓
1st Advanced rigger				✓	
2nd Advanced rigger.				✓	
Crane operator					✓

For non-team members, or persons not directly involved in the lifting procedure must either be refused entry to the work area or be inducted on the lifting plan as well as the risk assessment.

2. Site procedures

- All documentation must be completed correctly and communicated to the team via a toolbox talk (TBT)/ safety brief.
- Before any lifting commences, on-site a risk assessment, lifting plan and equipment pre-use inspection must be done and a drop zone must be established.
- All riggers on site must participate in the TBT conducted by the supervisor before work commences. Should an additional technician or any other person join the work site during lifting, the technician must be inducted by the supervisor.

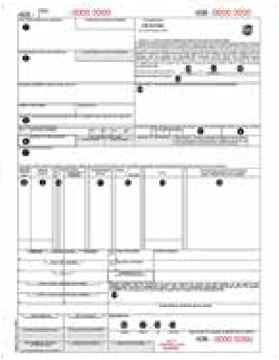


3. Load calculations

In order to determine the correct rigging procedure and equipment for the job at hand, an accurate load calculation will need to be conducted to ensure that the load required to be lifted is within the W.L.L. of the rigging equipment.

Two ways are generally used to determine the weight of a load:

- Load cell
- Waybill



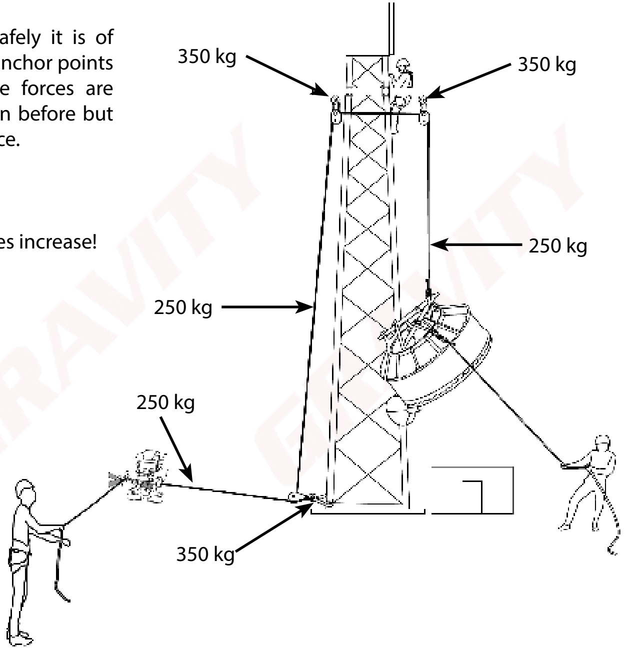
Manufacturers and companies can test one load with all its brackets and send a notice to all riggers informing them about the weight of that specific load for future applications.

Ensure that all additional weight is also accounted for when determining the load to be lifted.

2. Forces on tower

In order for a lift to be conducted safely it is of utmost importance that the forces on anchor points is understood and allowed for. These forces are affected by the angle of the lift as seen before but any deviation may also increase the force.

Remember: Forces increase as the angles increase!



3. Force calculations

Always consider the additional weight added to the load by factors such as taglines and wind.

Below follows a brief summary of calculations that can be used to determine the extra added weight.

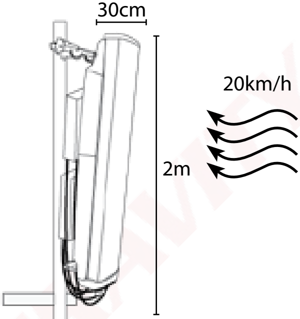
Wind loading:

Wind blowing against the surface area of the load can add considerable weight which in turn must then be counteracted by the taglines adding more force on the system and anchor points.

A simple formula exists to determine the weight added to a load due to wind:

F = APCd
F = Wind loading in kg
A = Surface area in m^2
P = $0.613 * v^2$ (where v = wind speed in m/s) in N/m^2
Cd = Drag coefficient, 1.4 for small rectangles, 0.8 for cylinders

Example:
A 2m x 0.3m antenna is being lifted in 20km/h winds, calculate the wind load.



Solution:
$A = 2m \times 0.3m = 0.6m^2$
$P = 0.613(20/3.6)^2 = 30{,}9\ N/m^2$
Cd = 1.4
Thus:
F = APCd
$F = (0.6m^2)(30.9N/m^2)(1.4)$
$= 18.919kg$
$\approx 19kg$

Tag lines:

Tag lines add a significant amount of weight to the lift and is dependant on the angles created by the tag lines with respect to the load as well as the angle created between the load and the tower.

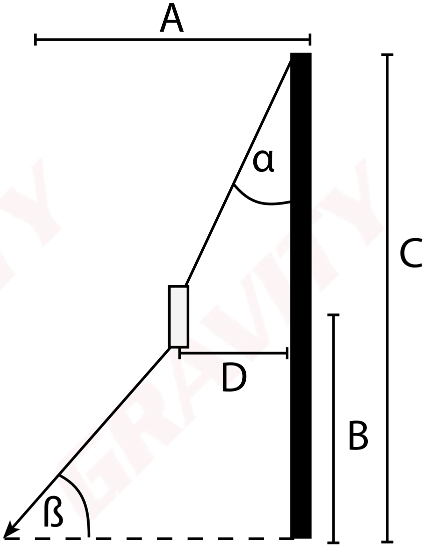
The angles are calculated as follows:

$$\alpha = \tan^{-1}(D/(C-B))$$

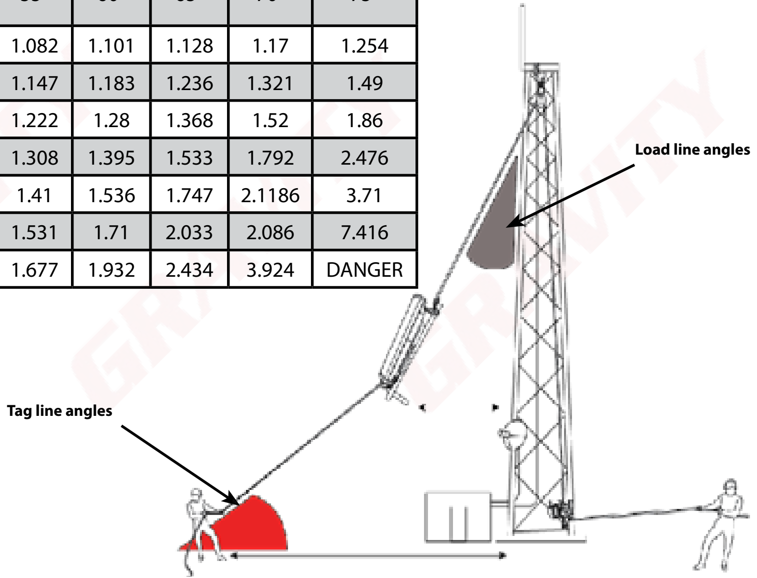
$$ß = \tan^{-1}(B/A-D)$$

The load factor table (given on the next page) is then used to determine the load factor, round the angles up to the next highest value and read the factor from the table. The new load is then calculated as follows:

Actual load = Load x LF



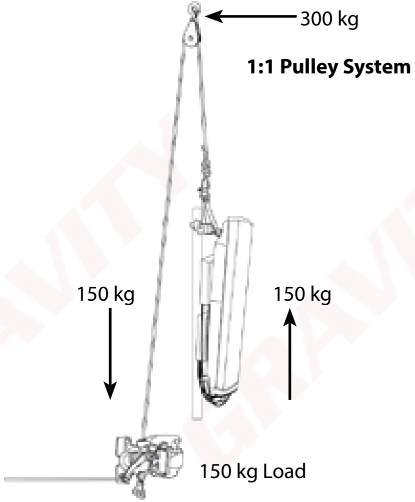
Load factors							
Load line angle	Tag line angle						
	45°	50°	55°	60°	65°	70°	75°
3	1.057	1.068	1.082	1.101	1.128	1.17	1.254
5	1.1	1.121	1.147	1.183	1.236	1.321	1.49
7	1.149	1.18	1.222	1.28	1.368	1.52	1.86
9	1.203	1.248	1.308	1.395	1.533	1.792	2.476
11	1.265	1.326	1.41	1.536	1.747	2.1186	3.71
13	1.334	1.416	1.531	1.71	2.033	2.086	7.416
15	1.414	1.521	1.677	1.932	2.434	3.924	DANGER



4. Mechanical Advantage Systems

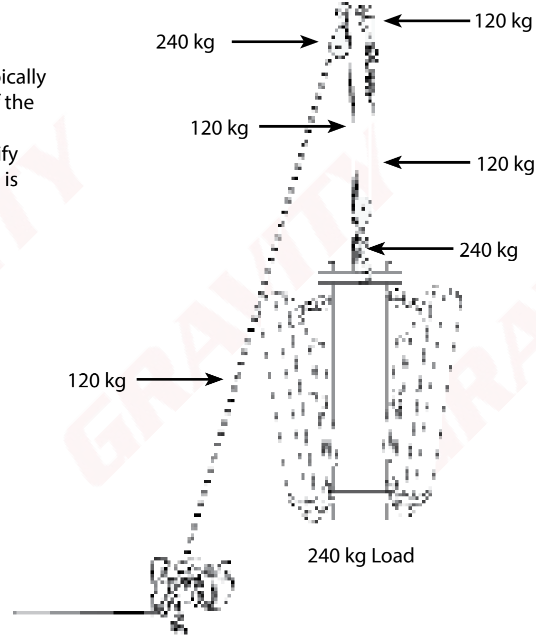
There are various mechanical advantage systems or pulley systems. All of them have their own advantages and disadvantages. Here are a few examples

When using a 1:1 Pulley system (as shown to the right) for a load between 1 and 40 kg, the typical team composition will consist of 2 persons, 1 supervisor and 1 hauler.



2:1 Pulley System

A 2:1 pulley system is typically used when the weight of the load is too much for a 1:1 system but does not justify the fact that a 3:1 system is quite a bit slower.



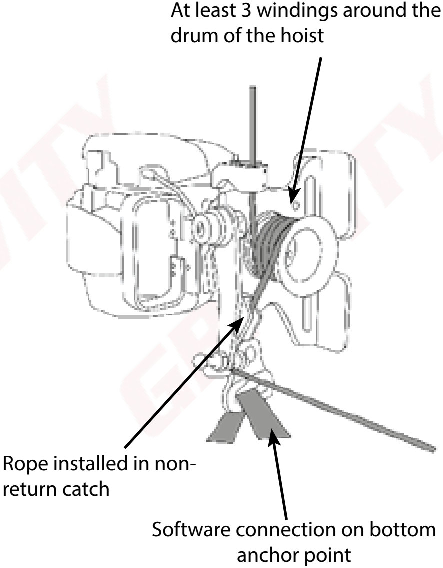
5. Capstan hoist

Installing the Capstan hoist:

- Ensure the tower leg is secure
- Ensure the chains of the bracket are secured as to prevent slippage
- Do not install the hoist directly underneath the top anchor point/gin pole

Before use ensure that:

- The rope is wound at least 3 times around the drum for loads between 40kg and 120kg. Wind the rope 4 times around the drum for loads above 120kg.
- The rope is installed in the non-return catch
- The bottom anchor point is securely fastened and tensioned
- Ensure there is no hardware through the bottom anchor point of the Eder winch
- Maintain tension on the pulling rope to ensure sufficient friction on the drum to allow the load to lift



Overwind

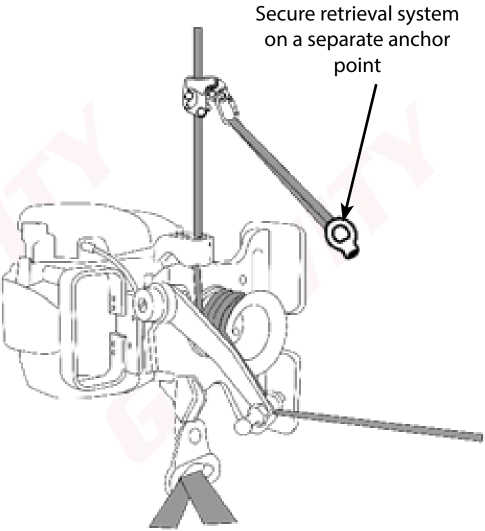
Overwinding occurs when you start using the hoist with a rope that is not properly pre - tensioned or if the weight of the load / amount of resistance is not enough.

How to avoid overwinding

- Properly Pre - tension the rope by pulling on the rope before winding it on to the hoist's drum.
- Ensure that the weight of the load is not less then 10% of the hoist's lifting capacity
- Lift at a steady rate/speed and keep your eyes on the drum and stop immediately if overwinding starts

Should the winch overwind

- Install a lifting sling on a separate anchor point
- Connect a Camp Goblin to the lifting sling using a karabiner
- Install the Camp Goblin on the load line
- Release the winch until the weight of load is fully suspended on the Camp Goblin
- Remove the windings from the winch drum and rewind the rope around the drum
- Lift the load using the winch until the Camp Goblin becomes slack
- Remove the Camp Goblin



6. Stop start process for Gravity Capstan hoists

• Eder 500 - 4 stroke

- Fill the Petrol tank to the level indicated on the tank.
- Fix the hoist in a working position.
- Bleed the primer about 3-7 times to fill the carburetor with fuel.
- Move the choke lever to the closed position.
- Switch the kill switch ON.
- Firmly take hold of the starter handle and pull until the motor starts.
- After the motor has warmed up while idling you can then slowly move the choke lever to the open position to allow the motor to run at its optimal revs.

- When starting a warm motor do not switch the choke lever to close.

- To stop the motor keep the rope slacked and ensure the throttle is set to idle.
- Allow the motor to cool by running it in idle for a few seconds.
- Turn the kill switch to the OFF position and the motor must stop.

• Eder 1200 - 2 stroke

- Fill the tank with the pre-mixed Petrol and 2 stroke oil to the indicated level of the tank.
- Fix the hoist in a working position.
- Bleed the primer about 1 or 2 times until the fuel is visible in the primer.
- Move the choke lever to the closed position.
- Firmly take hold of the starter handle and pull starter once.
- Move the choke lever to the open position.
- Switch the kill switch ON.
- Push the decompression button down.
- Firmly take hold of the starter handle and pull starter once.

- When starting a warm motor do not switch the choke lever to close.
- Switch the kill switch ON.
- Push the decompression button down.
- Firmly take hold of the starter handle and pull starter once.

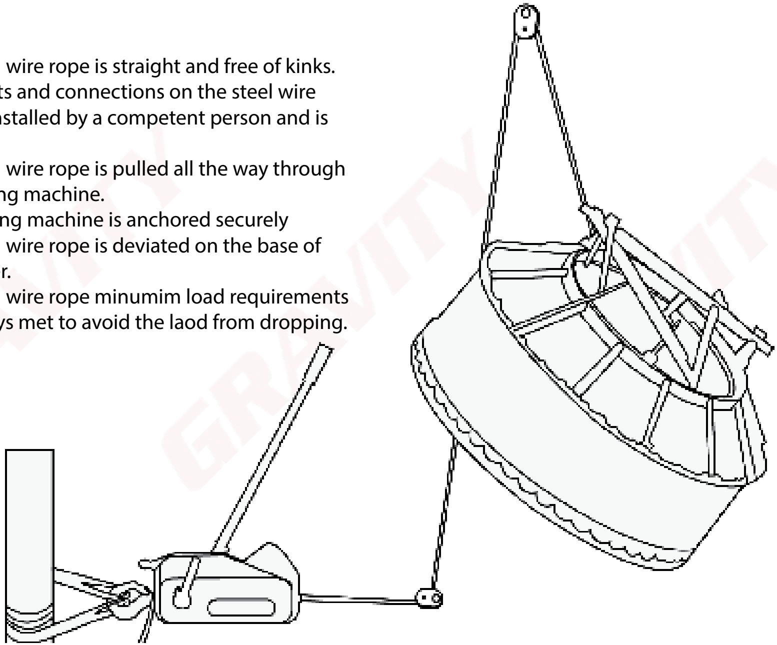
- To stop the motor keep the rope slacked and ensure the throttle is set to idle.
- Allow the motor to cool by running it in idle for a few seconds.
- Turn the kill switch to the OFF position and the motor must stop.

- As emergency kill switch to switch of the hoist, Push the decompression button down.

7. Tirfor

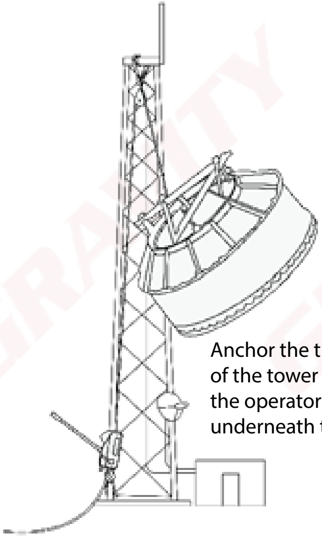
Ensure that:

- The steel wire rope is straight and free of kinks.
- All eyelets and connections on the steel wire rope is installed by a competent person and is secure.
- The steel wire rope is pulled all the way through the pulling machine.
- The pulling machine is anchored securely
- The steel wire rope is deviated on the base of the tower.
- The steel wire rope minumim load requirements are always met to avoid the laod from dropping.

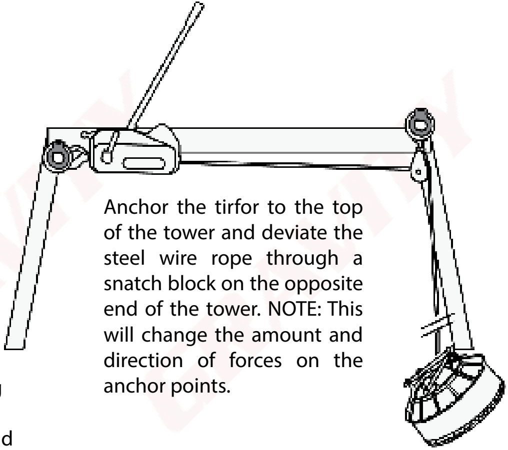


Alternative installations

Should there be limited anchor points on site or a restrictions on the length of the steel wire rope on site as shown below.



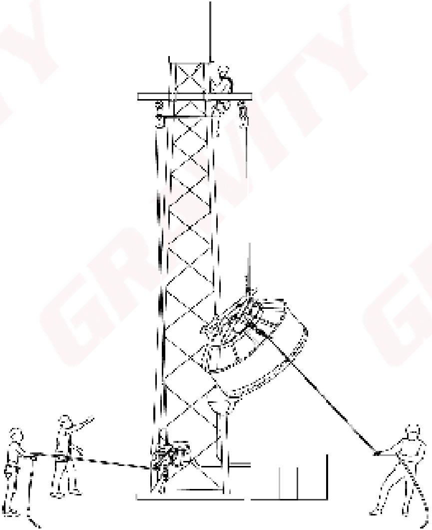
Anchor the tirfor to the leg of the tower ensuring that the operator does not stand underneath the load.



Anchor the tirfor to the top of the tower and deviate the steel wire rope through a snatch block on the opposite end of the tower. NOTE: This will change the amount and direction of forces on the anchor points.

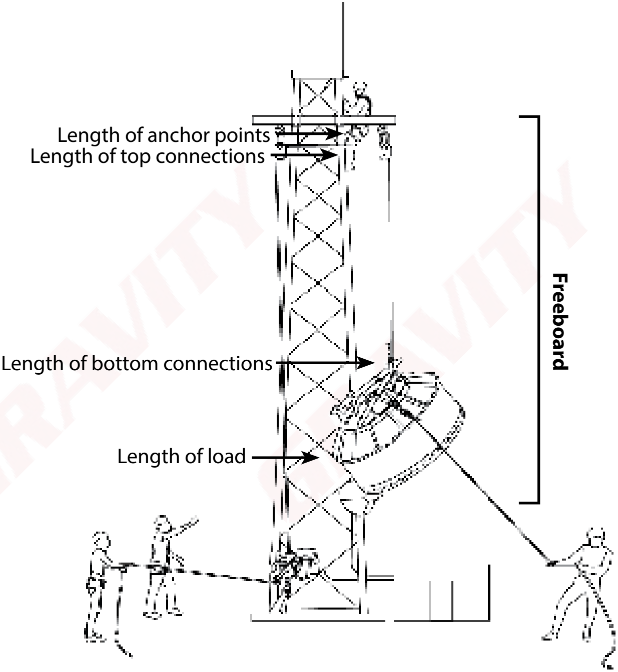
8. Rigging with a Gin-pole

1. Ensure that gin pole is lifted up to height with relevant rope rigging principles.
2. Install gin pole as per manufacturers instructions.
3. Ensure gin pole and tower is of sufficient strength for imposed loads.



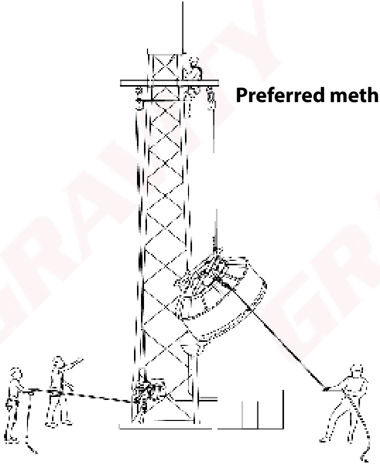
8. Freeboard

Freeboard refers to the minimum amount of space a rigger would need to lift a load to the desired height. The freeboard calculations are made as follows:

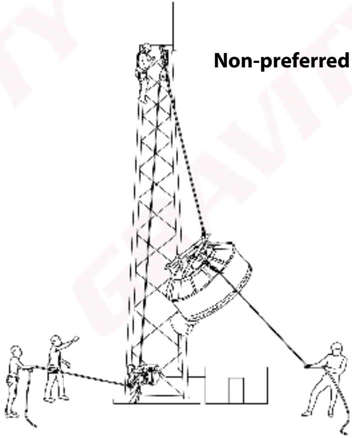


9. Lifting loads up to 250kg:

- Loads up to 250kg may be lifted with a 1:1 pulley system.
- When using a gin-pole, deviate the rope at the top in order to prevent the winch operator from standing underneath the load whilst lifting



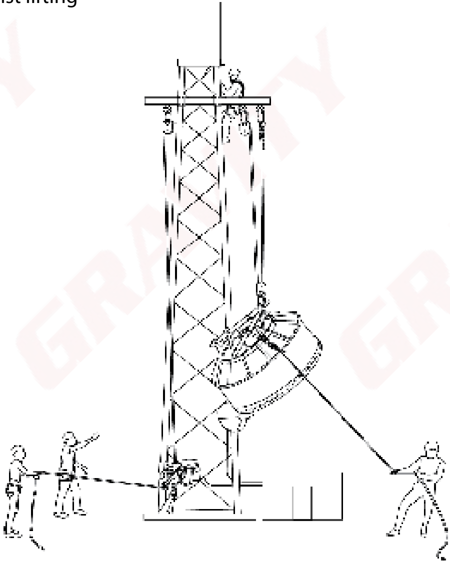
Preferred method



Non-preferred method

10. Lifting loads up to 500kg:

- Loads up to 500kg may be lifted with a 2:1 pulley system.
- When using a gin-pole, deviate the rope at the top in order to prevent the winch operator from standing underneath the load whilst lifting



11. Mounting the load to the structure

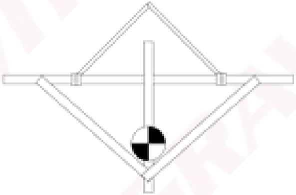
- All mounting brackets must be securely fastened and that there are no signs of over-torquing of the brackets that can cause damage to the structure.
- Once the load has been securely fastened the Lifting Supervisor should indicate that the lifting equipment can be removed.
- Note: Always pan the antenna or dish with the lifting equipment still secured to the load.

12. When lowering the load

- Ensure that the non-return system is secured to the load and structure prior to releasing the load from the structure.
- Ensure that the rope is tightened to the right tension so as to not allow the load to drop and shock load the system when releasing the load from the structure.
- Ensure that the lowering route is free from of obstacles.
- Ensure that when lowering the load does not pass over any obstacles which may cause any unnecessary friction.
- Exclusion zones must be maintained.

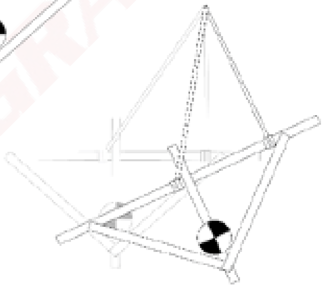
H. Slinging principles

1. Ensure that the slinging is done in such a way that the load is as secure in the air as it would be on the ground.
2. The slinging method should be adequate to the type of load, with attachment point to the load and the equipment.
3. The weight of the load should not exceed the WLL of the slinging gear.
4. The load must not damage the slinging gear and the slinging gear must not damage the load.
5. Know the weight of your load.
6. Consider the centre of gravity of the load.

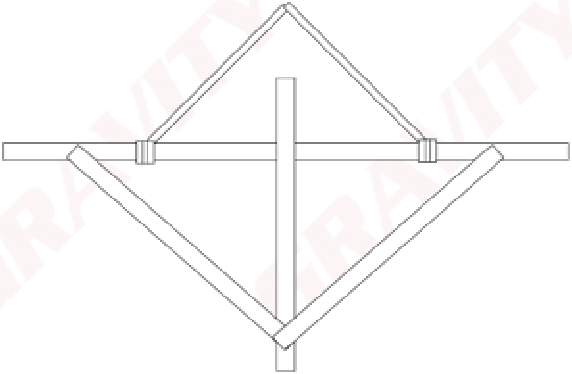


1. Centre of gravity:

- Very careful consideration has to be given to the centre of gravity of a load. This has to be considered on the horizontal and vertical axis.
- The sling lengths might have to be adjusted so that the centre of gravity is under the hook with the load level.
- When this adjustment is carried out, calculations will be required.

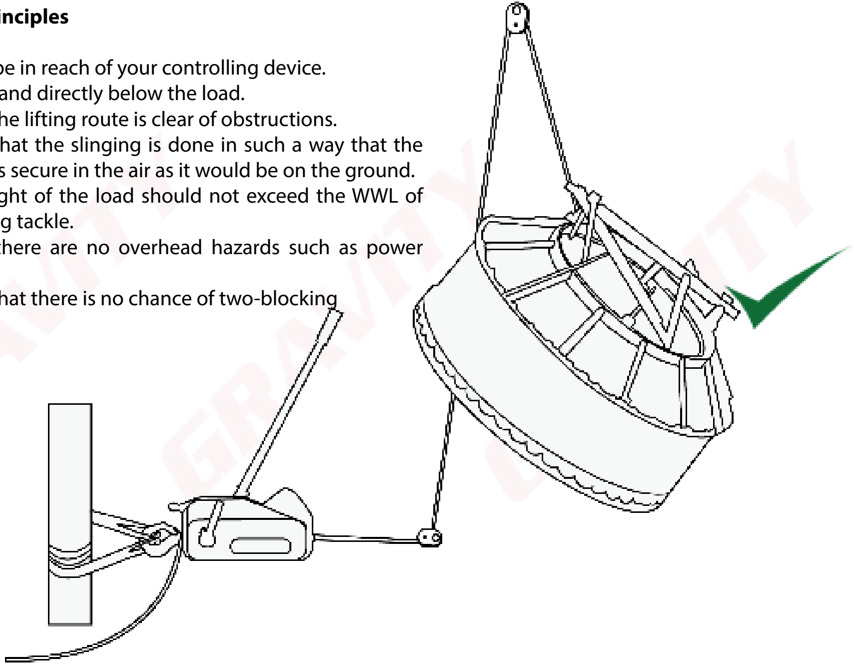


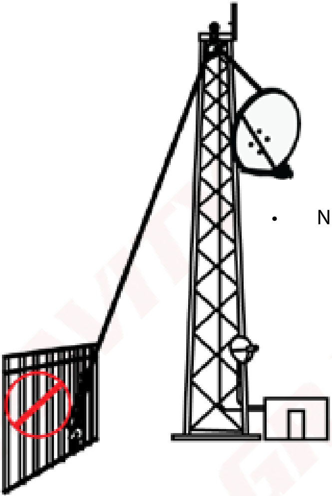
Examples of slinging methods:
- Horizontal orientation, and
- Vertical orientation.



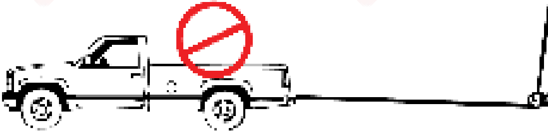
I. Rigging principles

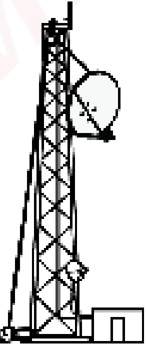
- Always be in reach of your controlling device.
- Never stand directly below the load.
- Ensure the lifting route is clear of obstructions.
- Ensure that the slinging is done in such a way that the load is as secure in the air as it would be on the ground.
- The weight of the load should not exceed the WWL of the lifting tackle.
- Ensure there are no overhead hazards such as power lines.
- Ensure that there is no chance of two-blocking



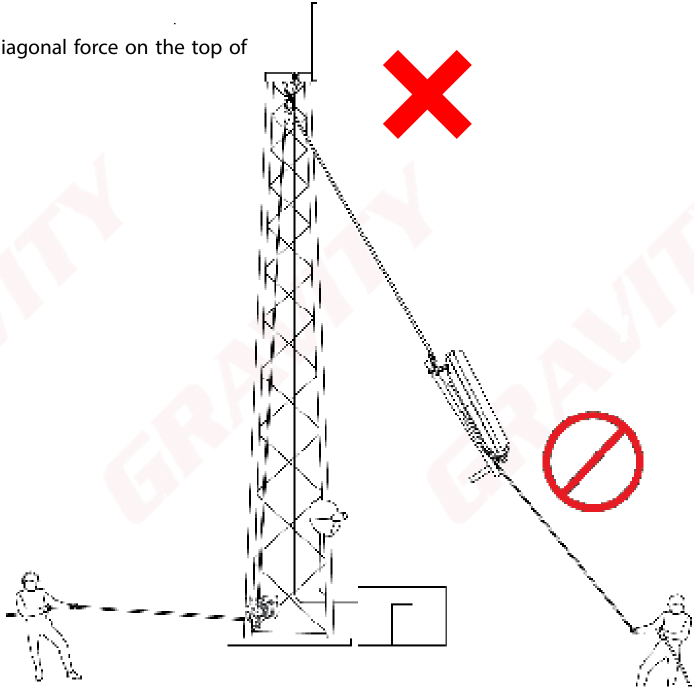


- NEVER rig from fencing!
- NEVER use vehicles for lifting!



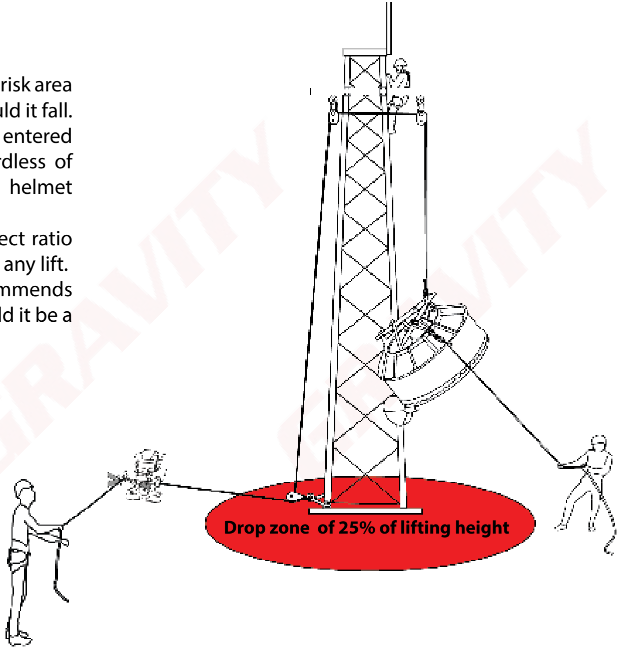


- Never put diagonal force on the top of the tower.

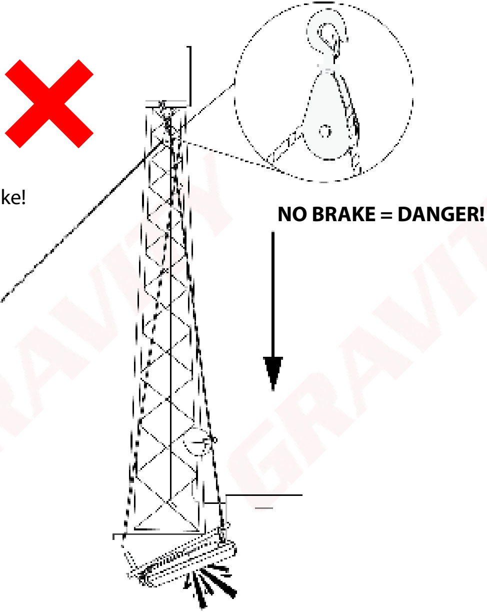


Drop zones

- The drop zone refers to the high risk area where equipment can land should it fall.
- The drop zone should not be entered during lifting operations, regardless of weather a person is wearing a helmet or not.
- Always ensure to have the correct ratio of **Drop zone** when performing any lift.
- The industry best practice recommends is 25% of the lifting height should it be a low and medium risk lift (4:1)

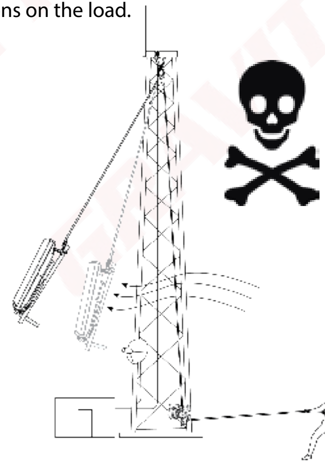


- ALWAYS use a device with a brake!



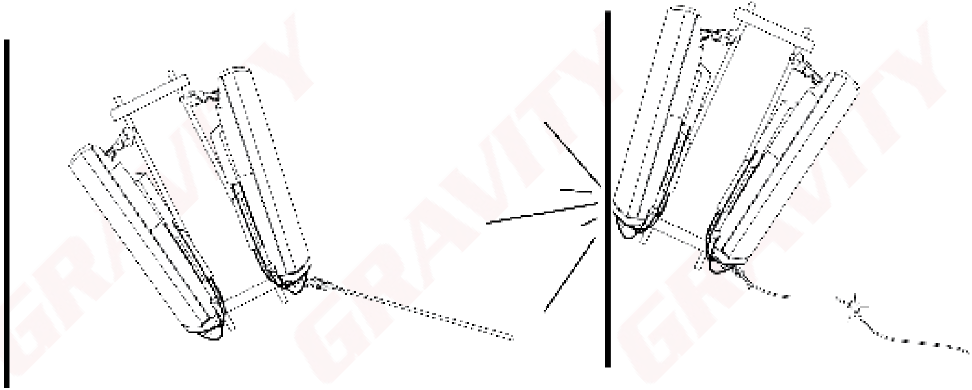
J. Taglines/Fly-lines:

- Taglines (Guide Ropes) are ropes attached to an object to steady, guide or secure the object.
- Double guide ropes are used where there might be high winds and the load needs to be kept steady or if a lifting operation that happens next to a building with glass panes or protruding edges or antennas.
- Careful calculations must be made when using taglines as it can add significant weight to the load.
- Where possible connect taglines and lifting ropes to the same connection point.
- Be careful to not apply excessive force in opposite directions on the load.



1. Mode of failure:

- The mode of failure refers to what will happen to the load should something in the rigging setup fail.



K. Hand signals used during lifting

Hand signals are an easy, effective and inexpensive way of communication. Because they are visual, they are almost unmistakable if kept simple. They are also universal, thus language barriers are eliminated.

Some guidelines to follow regarding the use of hand signals:

- All lifts should employ a signalman using standard hand signals.
- All signalmen must be qualified.
- The signalman must always be in clear view of the lifting operator.
- Signalman must always maintain visual contact of the load and the hoisting equipment.
- The lifting supervisor must act as the safety watch to keep the lift site free of unauthorized personnel and must stop should any unsafe activity arise.
- There should be only one designated signalman for each lift.
- The signaller should wear distinctive clothing.

Basic hand signals


Lift Load


Lower Load


Lower Load Slowly (Pre-Lift)


Lift Load Slowly (Pre-Lift)

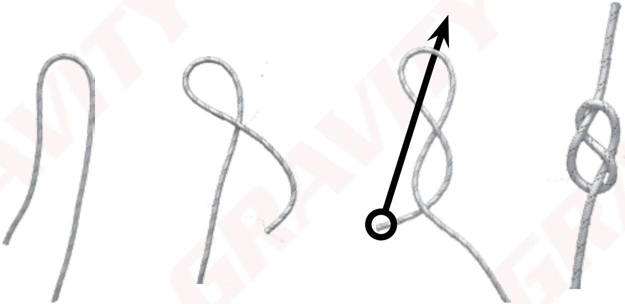

Stop


Dog Everything
(Pause until further notice)

L. Knots

- All knots weaken and decrease the length of the rope.
- All knots must be dressed properly; an incorrectly dressed knot can be up to 50% weaker than a knot dressed correctly.

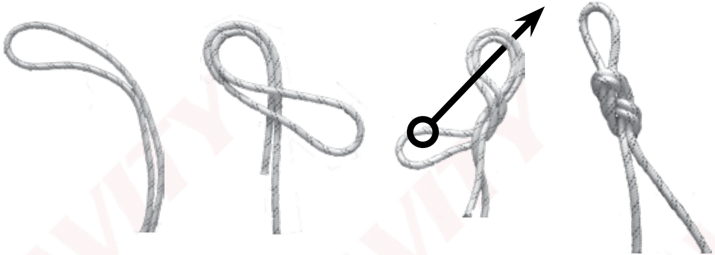
1. Figure 8 stopper knot



This knot is made in the end of a rope to stop people and equipment from being able of move off the end of the rope.

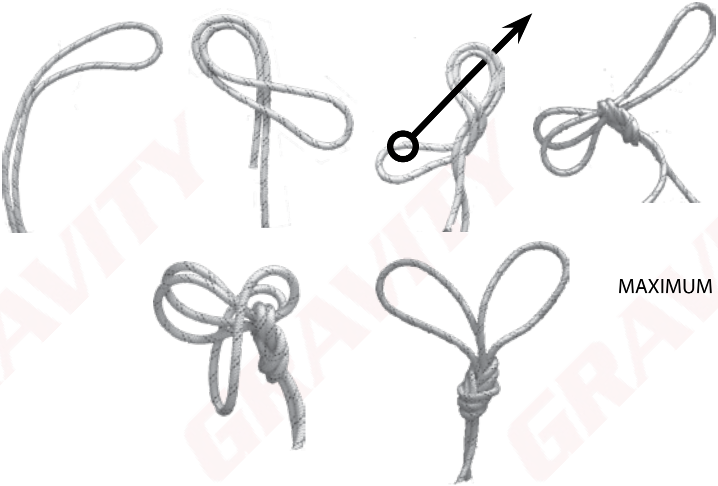
The tail must be longer than your fist, but shorter than your forearm.

2. Figure of 8 on a bight



This knot is used to connect the rope to a single anchor point or device.

3. Double loop figure 8



MAXIMUM ANGLE = 90°

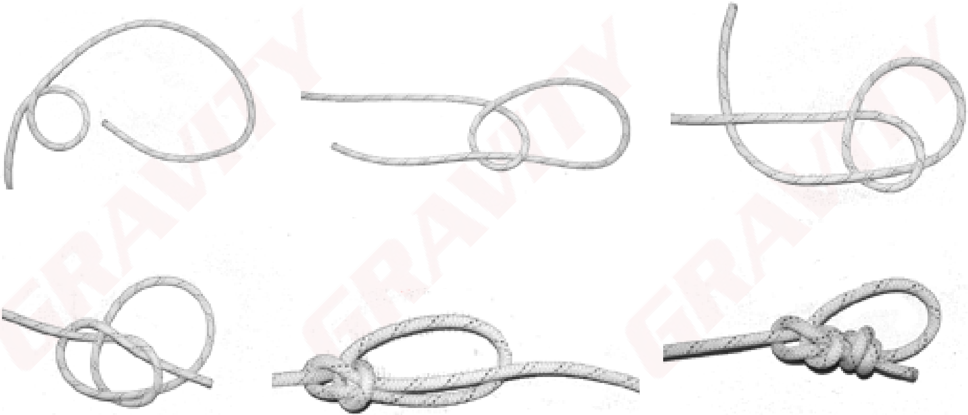
This knot is used to connect a rope to two separate anchor points and is able too adjust to compensate for center of gravity balancing.

4. Figure 9 on a bight

Use: Similar reasons as the Figure 8 on a Bight, however the Figure 9 is better suited when lifting heavier objects as it unties more easily after it has been loaded.

5. Bowline

Use: A knot that can be used like the Figure 9 for lifting heavier objects as it unties very easily after being loaded and is very quick to tie. It must, however, be tied off with a Fisherman knot.

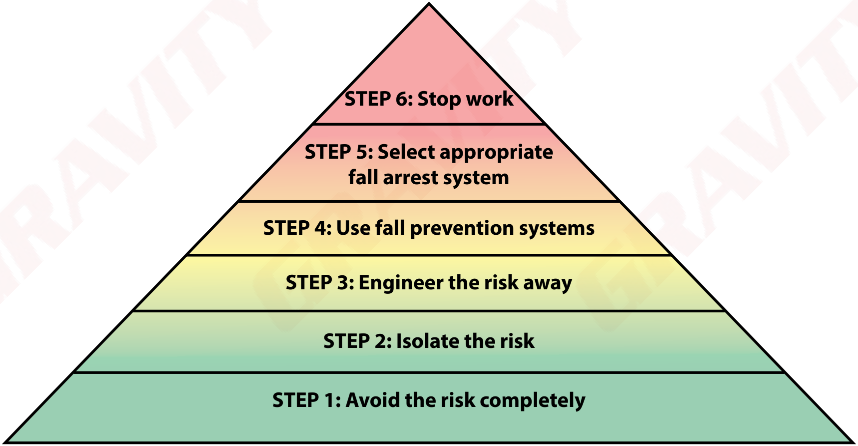


A Bowline tied off with a Barrel/ Fisherman knot.

M. Risk assessment

A risk assessment must be conducted by the site supervisor on site, daily, before work starts in order to ensure that all risks on site are controlled and mitigated.

Below is a flowchart depicting the process flow for risk mitigation



Adverse weather conditions:

No rigging or work-at-height shall be done during any inclement weather conditions. Should the situation change whilst at height, all technicians must safely descend to the ground and the Site Supervisor must report back immediately to the Project Manager.

Examples of inclement weather conditions:

- Rain
- Lightning
- Extremely cold weather
- Extremely hot weather
- Sleet or snow
- High or gusty wind (keep in mind shape and weight of load)
- Poor visibility (fog or smog)

Beaufort wind scale

Description	Wind speed	Appearance of wind effects	
		On a tree	On land
Calm	< 1 km/h	Still	Smoke rises vertically
Light	6 – 11 km/h	Leaves rustle	Wind felt on face, vanes begin to move
Moderate/Strong (Caution)	29 – 38 km/h	Small trees and leaves begin to sway	Flags fully extended
Strong	38 – 49 km/h	Larger branches begin to shake	Whistling in wires, um-brella becomes difficult to control
Gale	62 – 74 km/h	Whole trees move and twigs break	Windblown dust and dirt

N. Planning for a lift

Before arriving on site, there are certain very important pieces of information that the rigger requires. These must either be obtained by:

- Requesting them from the client
- Conducting a site visit

Factors to consider before completing lifting plan:

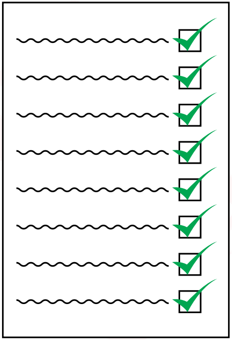
- All workers on site are medically fit and competent as per site and task
- Weather conditions allow for lift to commence uninterrupted
- Anchor points of sufficient strength for imposed loads - remember the doubling of loads on pulleys
- Correct equipment for lift (long enough rope/steel wire rope)
- Weight of the load has been considered with regards to all equipment and anchors
- Anchor angles have been considered
- Drop zones can be established and maintained
- Freeboard
- Loads imposed on tower to be approved by tower owner/engineer based on current load and new imposed load.

Failure to plan and prepare for a lift will lead to not all risks being managed which could lead to serious injury or death.

Lifting Plan

A lifting plan is a document specifying the how and by whom the lift will be completed. This must be completed by the Site Supervisor on site before work starts, and must include the following:

- The site name/address
- An approved work permit
- The system to be used (with back-up, taglines etc)
- The team composition
- Description of the lift
- The type and weight of the load
- Drop zones and exclusion zones
- Anchor points to be used
- Weather conditions
- The expected duration of the lift
- The supervisor’s details



If at any point alterations have to be made to the lifting operation which were not indicated on the lifting plan, it must be documented and signed by the supervisor before continuing with the lift.

Detailed lifting plan

Site details:

Site Name : ____________________ *Site No :* ____________________

Site Supervisor Name : ____________________

Contact Number : ____________________ *Date :* ____________________

No. of workers on site : ____________________

Lift details : ____________________

____

____

Type of anchor pionts with standard and rating:

____

Methods of communication during the lift : **HAND SIGNALS**

TWO WAY RADIO ☐ **VERBAL** ☐

Admin requirements : *(These requirements are the responsibility of the Site Supervisor as mentioned avove. e.g.Training, PPE, Inspection etc)*

1. ______________________ 2. ______________________
3. ______________________ 4. ______________________
5. ______________________ 6. ______________________

Equipment requirements : *(Include SWL or WLL per item)*

Rope rigging equipment	Qty	Mechanical lifting equipment	Qty

PPE	Qty	Tools	Qty

Lift 1 load discription : ______________

Weight of the load in Kg : ________________________ *Shape of the load :* ___________________________

Pulley System to be used : ________________________ *Estimate tag line weight* ______________________

All up weight : ________________________________ *Top anchor load in Kg :* ________________________

Height of the lift in meters : ______________________________________ *Lift* ☐ *or Lift & lower* ☐

Team compesition for lift 1: ____

Draw a detailed plan with the type of system that will be used for the lifting of the load. Indicate the weight of the load and the weight that will be placed on the top and bottom anchor once the lift starts while considering the weight that the tagline forces will add.

Lift 1 sign off:

Lift planner and supervisor: I confirm that I have planned this lift in accordance with IBP procedures and accept the responsibilities of my position.

Initials and Surname	Signature	Date

Change notes:

(Should anything happen during the lift causing a deviation from the lifting plan, please note it here, this includes: the use of lifting equipment and relevant accessories)

Change notes: sign off:

Lift planner and supervisor: I confirm that I have planned this lift in accordance with IBP procedures and accept the responsibilities of my position.

Initials and Surname	Signature	Date

Pre-operational checks:

- [ ] NO VEHICLES ARE TO BE USED FOR LIFTING
- [ ] THE EFFECT THAT THE WIND HAS ON THE LOAD HAS BEEN CONSIDERED
- [ ] THE EFFECT OF THE TAG LINES HAS ON THE LOAD HAS BEEN CONSIDERED
- [ ] ALL FORCE WILL BE APPLIED AT BASE OF TOWER SHOULD THE HAULING LINE BE DEVIATED
- [ ] NO ONE TO STAND UNDERNEATH LOAD
- [ ] NO BRAIDED ROPES ARE TO BE USED
- [ ] NO NYLON (SKI) ROPES ARE TO BE USED
- [ ] RIG FROM ANCHORS OF SUITABLE STRENGTH (NO FENCES, VEHICLES ETC)
- [ ] THE CENTER OF GRAVITY OF THE LOAD HAS BEEN CONSIDERED
- [ ] THE LOAD HAS BEEN CORRECTLY SLUNG
- [ ] ONLY EQUIPMENT THAT IS SAFE FOR USE AND FIT FOR PURPOSE IS TO BE USED
- [ ] THE LIFTING/LOWERING ROUTE IS CLEAR OF OBSTRUCTIONS
- [ ] THE LOAD DOES NOT EXCEED WLL OF EQUIPMENT
- [ ] THE BANKSMAN IS VISIBLE AT ALL TIMES
- [ ] WEATHER CONDITIONS HAVE BEEN CONSIDERED
- [ ] NO WORKER TO CLIMB WITH LOAD (excl. tools) CONNECTED TO THEMSELVES
- [ ] NO UNDERGROUND SERVICES, OVERHEAD POWERLINES ETC
- [ ] STABILITY OF STRUCTURE AND LIFTING AREA CONSIDERED
- [ ] ALL WORKERS ON SITE HAVE RELEVANT COMPETENCIES
- [ ] RELEVANT PERMISSIONS HAVE BEEN OBTAINED

Agree and confirm:

I, the undersigned, agree that the I understand and accept the plan as written above.

Name and Surname	Designation	Signature
1.		
2.		
3.		
4.		
5.		
6.		
7.		
8.		
9.		
10.		
11.		
12.		
13.		
14.		
15.		

THE WAY FORWARD

Congratulations on completing Gravity Training's Mechanical Lifting course.

If found not yet competent (NYC), you will need more training before you can be reassessed. Please do not be discouraged. Not everyone passes on their first attempt. Some learners need a bit more experience to obtain the required knowledge and skills. You will receive comprehensive feedback on your performance and be informed on how to proceed.

At Gravity, we care about the professional development of our clients. Should you wish to take the next step, consider enrolling for some of our other courses. See The Way Forward and the Gravity Skills and Development Map or consult your facilitator/assessor.

GRAVITY COURSES